

Differentiation

Definition of Derivative

Definition: Let $f : [a, b] \rightarrow \mathbb{R}$ and $c \in (a, b)$. The derivative of f at c is defined as

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h},$$

if the limit exists.

Geometrical Interpretation

- The derivative at a point gives the slope of the tangent line.
- Positive derivative $\Rightarrow$ increasing function.
- Negative derivative $\Rightarrow$ decreasing function.

Theorem

Theorem:

Let I be an interval and a function $f : I \rightarrow \mathbb{R}$ be differentiable at a point $c \in I$. Then f is continuous at c .

Proof

For all $x \in I$, $x \neq c$,

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} (x - c).$$

Since f is differentiable at c ,

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = f'(c),$$

which is finite.

Also,

$$\lim_{x \rightarrow c} (x - c) = 0.$$

Therefore,

$$\lim_{x \rightarrow c} (f(x) - f(c)) = f'(c) \cdot 0 = 0,$$

so

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Hence f is continuous at c .

The continuity of f at $c \in I$ does **not** ensure differentiability at c .

Example: Let $f(x) = |x|$, $x \in \mathbb{R}$.

- At $x = 0$, $f(0) = 0 \Rightarrow f$ is continuous at 0.
- Right-hand derivative:

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1.$$

- Left-hand derivative:

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1.$$

Since right and left derivatives are not equal, f is not differentiable at 0.

Rules of Differentiation

If u, v are differentiable, then:

$$(u + v)' = u' + v', \quad (uv)' = u'v + uv',$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \quad (cu)' = cu'.$$

Sign of Derivative and Monotonicity

Definitions

A function f is:

- **Increasing at c :** $f(x) < f(c)$ for $x < c$ and $f(x) > f(c)$ for $x > c$
- **Decreasing at c :** $f(x) > f(c)$ for $x < c$ and $f(x) < f(c)$ for $x > c$

Theorem

If f differentiable at c :

- $f'(c) > 0 \Rightarrow f$ increasing at c
- $f'(c) < 0 \Rightarrow f$ decreasing at c

Important Notes

- Increasing at point $\nRightarrow$ positive derivative
- Derivative can be zero at increasing/decreasing points
- Local behavior doesn't guarantee global monotonicity

Darboux's Theorem: Intermediate Value Property

Theorem 9.4.1 (Darboux)

If f differentiable on $[a, b]$ and $f'(a) \neq f'(b)$, then for any k between $f'(a)$ and $f'(b)$, there exists $c \in (a, b)$ such that $f'(c) = k$

Key Implications

- Derivatives have the intermediate value property
- Derived functions cannot have jump discontinuities
- $f'(I)$ is an interval when I is an interval

Application

Shows that functions with jump discontinuities (like the example with $f(x) = 0$ on $[-1, 0]$ and $f(x) = 1$ on $(0, 1]$) cannot be derivatives of any function

Mean Value Theorems

Theorem

Rolle's Theorem: If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.

Theorem

Lagrange's Mean Value Theorem: If f is continuous on $[a, b]$ and differentiable on (a, b) , then

$$\exists c \in (a, b) : f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Corollary: Rolle's Theorem for Polynomials

Theorem

If a polynomial function f has at least two real roots, then between any two real roots there exists at least one real root of the derived polynomial function f' .

Proof.

Let α, β be two real roots of the polynomial function f , with $\alpha < \beta$.

Then:

- (i) f is continuous on $[\alpha, \beta]$,
- (ii) f is differentiable on (α, β) , and
- (iii) $f(\alpha) = f(\beta) = 0$.

Therefore, by Rolle's theorem, there exists at least one real number $\xi \in (\alpha, \beta)$ such that $f'(\xi) = 0$.

Example

Consider the polynomial function:

$$f(x) = x^3 - 6x^2 + 11x - 6$$

Let's factor the polynomial:

$$f(x) = (x - 1)(x - 2)(x - 3)$$

The real roots are:

$$\alpha = 1, \quad \beta = 2, \quad \gamma = 3$$

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Between any two consecutive roots, say $\alpha = 1$ and $\beta = 2$:

- ① f is continuous on $[1, 2]$ (all polynomials are continuous everywhere)
- ② f is differentiable on $(1, 2)$ (all polynomials are differentiable everywhere)
- ③ $f(1) = f(2) = 0$

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- ① f is continuous on $[1, 2]$ (all polynomials are continuous everywhere)
- ② f is differentiable on $(1, 2)$ (all polynomials are differentiable everywhere)
- ③ $f(1) = f(2) = 0$

By Rolle's Theorem, there exists $\xi_1 \in (1, 2)$ such that:

$$f'(\xi_1) = 0$$

cont....

Similarly, between $\beta = 2$ and $\gamma = 3$, there exists $\xi_2 \in (2, 3)$ such that:

$$f'(\xi_2) = 0$$

The derivative is:

$$f'(x) = 3x^2 - 12x + 11$$

cont....

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Solving $f'(x) = 0$:

$$x = \frac{12 \pm \sqrt{144 - 132}}{6} = \frac{12 \pm \sqrt{12}}{6} = \frac{12 \pm 2\sqrt{3}}{6} = 2 \pm \frac{\sqrt{3}}{3}$$

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Approximating:

$$\xi_1 = 2 - \frac{\sqrt{3}}{3} \approx 1.422 \in (1, 2)$$

$$\xi_2 = 2 + \frac{\sqrt{3}}{3} \approx 2.577 \in (2, 3)$$

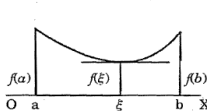
Cauchy's Mean Value Theorem

If f, g are continuous on $[a, b]$, differentiable on (a, b) , and $g'(x) \neq 0$ in (a, b) , then $\exists c \in (a, b)$ such that

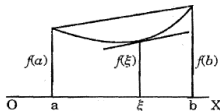
$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Geometrical interpretation.

If a function f has a graph which is a continuous curve on the interval $[a, b]$ and the curve has a tangent at every point on it with abscissa between a and b then there exists a point ξ in (a, b) such that the tangent to the curve at $(\xi, f(\xi))$ is parallel to the line segment joining the points $(a, f(a))$ and $(b, f(b))$.



Rolle's theorem



Mean value theorem

Remark.

Rolle's theorem is a particular case of Mean value theorem. If $f(a) = f(b)$ holds in addition to the two conditions of Mean value theorem, then $f(b) - f(a) = 0$ and consequently $f'(\xi) = 0$.

In the particular case, the geometrical interpretation is that there is a point $(\xi, f(\xi))$ on the curve, the tangent at which is parallel to the x -axis.

Theorem 9.5.3. Let a function $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f'(x) = 0$ for all $x \in (a, b)$ then f is a constant on $[a, b]$.

Proof. Let $x_1, x_2 \in [a, b]$ and $a \leq x_1 < x_2 \leq b$.

Then f is continuous on $[x_1, x_2]$, and f is differentiable on (x_1, x_2) .

By the Mean value theorem there exists a point ξ in (x_1, x_2) such that $f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$.

But $f'(\xi) = 0$, by hypothesis. Therefore $f(x_2) = f(x_1)$.

Since x_1 and x_2 are arbitrary points in $[a, b]$, f is a constant on $[a, b]$.

Corollary. Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be both continuous on $[a, b]$ and they are both differentiable on (a, b) . If $f'(x) = g'(x)$ for all $x \in (a, b)$ then $f = g + c$, where $c \in \mathbb{R}$ is a constant.

Theorem 9.5.4. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $f'(x) \geq 0$ for all $x \in (a, b)$. Then f is a monotone increasing function on $[a, b]$.

Remark

Rolle's theorem is a particular case of Mean Value Theorem. If $f(a) = f(b)$, then $f(b) - f(a) = 0$ and consequently $f'(\xi) = 0$.

Key Theorems from MVT

Theorem (Constant Function Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.

Proof.

For any $x_1, x_2 \in [a, b]$ with $x_1 < x_2$, by MVT:

$$f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = 0 \Rightarrow f(x_2) = f(x_1)$$

Thus, f is constant. □

Corollary

If $f'(x) = g'(x)$ for all $x \in (a, b)$, then $f = g + c$ for some constant $c \in \mathbb{R}$.

Monotonicity Theorems

Theorem (Increasing Function Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $f'(x) \geq 0$ for all $x \in (a, b)$. Then f is monotone increasing on $[a, b]$.

Proof.

For $x_1 < x_2$ in $[a, b]$, by MVT:

$$f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \geq 0 \Rightarrow f(x_2) \geq f(x_1)$$



Theorem (Decreasing Function Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $f'(x) \leq 0$ for all $x \in (a, b)$. Then f is monotone decreasing on $[a, b]$.

Note

Worked Examples

Example (1)

Prove that $\frac{x}{2} < \sin x$ for $0 < x < \frac{\pi}{2}$.

Proof.

Let $f(x) = \frac{\sin x}{x}$ for $0 < x \leq \frac{\pi}{2}$.

f is continuous on $(0, \frac{\pi}{2}]$ and differentiable.

$f'(x) = \frac{x \cos x - \sin x}{x^2} < 0$ for $0 < x < \frac{\pi}{2}$ (since $x < \tan x$).

Thus f is strictly decreasing.

Since $f(\frac{\pi}{2}) = \frac{2}{\pi}$, we have $f(x) > \frac{2}{\pi}$ for $0 < x < \frac{\pi}{2}$,

i.e., $\frac{\sin x}{x} > \frac{2}{\pi} \Rightarrow \sin x > \frac{2x}{\pi}$. □

Worked Examples (Continued)

Example (2)

Prove that $\frac{x}{2} < \log(1+x) < x$ for all $x > 0$.

Proof.

Part 1: Let $f(x) = \log(1+x) - \frac{x}{2}$, $x \geq 0$.

$$f'(x) = \frac{1}{1+x} - \frac{1}{2} = \frac{1-x}{2(1+x)^2} > 0 \text{ for } 0 < x < 1$$

Thus f is increasing, so $f(x) > f(0) = 0$ $\log(1+x) > \frac{x}{2}$.

Part 2: Let $g(x) = x - \log(1+x)$, $x \geq 0$.

$$g'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0 \text{ for } x > 0$$

Thus g is increasing, so $g(x) > g(0) = 0$ $x > \log(1+x)$. □

Higher Order Derivatives

Definition:

$$f''(x) = \frac{d}{dx}(f'(x)), \quad f^{(n)}(x) = \frac{d^n}{dx^n} f(x).$$

Example: For $f(x) = e^x$, we have $f^{(n)}(x) = e^x$ for all n .

Taylor's Theorem

Theorem: Let f be $(n + 1)$ times differentiable on an interval containing a and x . Then

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x),$$

where the remainder term is

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n + 1)!}(x - a)^{n+1}, \quad \text{for some } c \text{ between } a \text{ and } x.$$

Taylor Polynomial

The polynomial part of degree n is called the **Taylor Polynomial**:

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

- Approximates $f(x)$ near $x = a$.
- The error is given by the remainder term $R_n(x)$.

Example: Expansion of e^x at $a = 0$

$$f(x) = e^x, \quad f^{(k)}(0) = 1 \quad \forall k$$

$$P_n(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}$$

$$R_n(x) = \frac{e^c}{(n+1)!} x^{n+1}, \quad c \in (0, x)$$

Thus,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Example 1: Expansion of $\sin x$ about $a = 0$

$$f(x) = \sin x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1}, \quad c \in (0, x)$$

Example 2: Expansion of $\cos x$ about $a = 0$

$$f(x) = \cos x$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Example 3: Expansion of $\ln(1 + x)$ about $a = 0$

$$f(x) = \ln(1 + x), \quad |x| < 1$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Example 4: Expansion of $(1 + x)^r$

$$(1 + x)^r = 1 + rx + \frac{r(r-1)}{2!}x^2 + \frac{r(r-1)(r-2)}{3!}x^3 + \dots, \quad |x| < 1$$

Example 6: Expansion of e^x about $a = 1$

$$f(x) = e^x$$

$$e^x = e \left[1 + (x - 1) + \frac{(x - 1)^2}{2!} + \frac{(x - 1)^3}{3!} + \dots \right]$$

Applications

- Approximating functions by polynomials.
- Numerical methods (Newton's method, integration, etc.).
- Error estimation in approximations.
- Basis for power series and analytic functions.

Definitions: Global and Local Extrema

Global (Absolute) Extrema

A function $f : I \rightarrow \mathbb{R}$ has:

- A **global maximum** at $c \in I$ if $f(c) \geq f(x)$ for all $x \in I$
- A **global minimum** at $c \in I$ if $f(c) \leq f(x)$ for all $x \in I$

Local (Relative) Extrema

A function $f : I \rightarrow \mathbb{R}$ has:

- A **local maximum** at $c \in I$ if $\exists \delta > 0$ such that $f(c) \geq f(x)$ for all $x \in N(c, \delta) \cap I$
- A **local minimum** at $c \in I$ if $\exists \delta > 0$ such that $f(c) \leq f(x)$ for all $x \in N(c, \delta) \cap I$

Note

If f has a local extremum at $c \in I$, then c is a global extremum point on $N(c, \delta) \cap I$ for some $\delta > 0$.

Necessary Condition for Local Extrema

Theorem

Let $f : I \rightarrow \mathbb{R}$ have a local extremum at an interior point c of I . If $f'(c)$ exists, then $f'(c) = 0$.

Proof.

We prove for local maximum case (local minimum is similar):

- Suppose $f'(c) > 0$: Then $\exists \delta > 0$ such that $\frac{f(x)-f(c)}{x-c} > 0$ for $x \in N'(c, \delta)$
- For $x \in (c, c + \delta)$: $x - c > 0 \Rightarrow f(x) > f(c) \leftarrow$ Contradiction!
- Suppose $f'(c) < 0$: Then $\exists \delta > 0$ such that $\frac{f(x)-f(c)}{x-c} < 0$ for $x \in N'(c, \delta)$
- For $x \in (c - \delta, c)$: $x - c < 0 \Rightarrow f(x) > f(c) \leftarrow$ Contradiction!
- Therefore, $f'(c) = 0$



Important Notes on Extrema

Corollary

If f has a local extremum at interior point c , then either:

- ① $f'(c)$ does not exist, OR
- ② $f'(c) = 0$

Note 1

A function may have extremum without being differentiable.

Example: $f(x) = |x|$ has local minimum at 0, but $f'(0)$ doesn't exist.

Note 2

$f'(c) = 0$ is necessary but not sufficient for extremum.

Example: $f(x) = x^3$ has $f'(0) = 0$ but no extremum at 0.

Note 3

Theorem only applies to interior points.

Example: $f(x) = x$ on $[0, 1]$ has maximum at 1 (endpoint) $f'(1) = 1 \neq 0$.

First Derivative Test for Extrema

Theorem

Let f be continuous on $I = [a, b]$ and c be an interior point. Let f be differentiable on (a, c) and (c, b) .

- ① If $f'(x) \geq 0$ on $(c - \delta, c)$ and $f'(x) \leq 0$ on $(c, c + \delta)$, then f has local maximum at c*
- ② If $f'(x) \leq 0$ on $(c - \delta, c)$ and $f'(x) \geq 0$ on $(c, c + \delta)$, then f has local minimum at c*
- ③ If $f'(x)$ has same sign on both sides, then no extremum at c*

Examples of First Derivative Test

Example (1)

$$f(x) = |x|, x \in \mathbb{R}$$

- $f'(x) < 0$ for $x \in (-\delta, 0)$
- $f'(x) > 0$ for $x \in (0, \delta)$
- Local minimum at 0

Example (2)

$$f(x) = |x - 1| + |x - 2|, x \in [0, 3]$$

- Not differentiable at 1 and 2
- $f'(x) < 0$ on $(1 - \delta, 1)$, $f'(x) = 0$ on $(1, 1 + \delta)$ Local min at 1
- $f'(x) = 0$ on $(2 - \delta, 2)$, $f'(x) > 0$ on $(2, 2 + \delta)$ Local min at 2

Example (3)

$$f(x) = (x - 1)^2(x - 3)^3, x \in \mathbb{R}$$

Higher Order Derivative Test

Theorem

Let $f : I \rightarrow \mathbb{R}$ and c be an interior point.

If $f'(c) = f''(c) = \dots = f^{(n-1)}(c) = 0$ and $f^{(n)}(c) \neq 0$, then:

- No extremum at c if n is odd
- Local maximum at c if n is even and $f^{(n)}(c) < 0$
- Local minimum at c if n is even and $f^{(n)}(c) > 0$